

JEE Mains 2019 Chapter wise Question Bank

Electrostatics - Questions

Q1

Three charges $+Q$, q , $+Q$ are placed respectively, at distance, $d/2$ and d from the origin, on the x -axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is:

- (1) $-Q/4$ (2) $+Q/2$
(3) $+Q/4$ (4) $-Q/2$

9 Jan Morning

Q2

For a uniformly charged ring of radius R , the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is:

- (1) $\frac{R}{\sqrt{5}}$ (2) $\frac{R}{\sqrt{2}}$
(3) R (4) $R\sqrt{2}$

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Q3

Two point charges q_1 ($\sqrt{10} \mu\text{C}$) and q_2 ($-25 \mu\text{C}$) are placed on the x -axis at $x = 1 \text{ m}$ and $x = 4 \text{ m}$ respectively. The electric field (in V/m) at a point $y = 3 \text{ m}$ on y -axis is,

$$\left[\text{take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2} \right]$$

- (1) $(63\hat{i} - 27\hat{j}) \times 10^2$
(2) $(-63\hat{i} + 27\hat{j}) \times 10^2$
(3) $(81\hat{i} - 81\hat{j}) \times 10^2$
(4) $(-81\hat{i} + 81\hat{j}) \times 10^2$

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Q4

Charge is distributed within a sphere of radius R with a volume charge density $p(r) = \frac{A}{r^2} e^{-2r/a}$ where A and a are constants. If Q is the total charge of this charge distribution, the radius R is:

$$(1) a \log \left(1 - \frac{Q}{2\pi a A} \right) \quad (2) \frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$$

$$(3) a \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right) \quad (4) \frac{a}{2} \log \left(1 - \frac{Q}{2\pi a A} \right)$$

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Q5

A charge Q is distributed over three concentric spherical shells of radii a, b, c ($a < b < c$) such that their surface charge densities are equal to one another.

The total potential at a point at distance r from their common centre, where $r < a$, would be:

$$(1) \frac{Q}{12\pi\epsilon_0} \frac{ab + bc + ca}{abc}$$

$$(2) \frac{Q(a^2 + b^2 + c^2)}{4\pi\epsilon_0(a^3 + b^3 + c^3)}$$

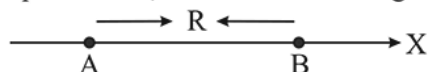
$$(3) \frac{Q}{4\pi\epsilon_0(a + b + c)}$$

$$(4) \frac{Q(a + b + c)}{4\pi\epsilon_0(a^2 + b^2 + c^2)}$$

10 Jan Morning

Q6

Two electric dipoles, A, B with respective dipole moments $\vec{d}_A = -4qa\hat{i}$ and $\vec{d}_B = -2qa\hat{i}$ are placed on the x-axis with a separation R, as shown in the figure



The distance from A at which both of them produce the same potential is:

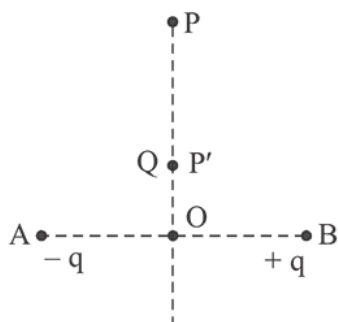
- (1) $\frac{R}{\sqrt{2}+1}$ (2) $\frac{\sqrt{2}R}{\sqrt{2}+1}$
 (3) $\frac{R}{\sqrt{2}-1}$ (4) $\frac{\sqrt{2}R}{\sqrt{2}-1}$

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Q7

Charges $-q$ and $+q$ located at A and B, respectively constitute an electric dipole. Distance $AB = 2a$, O is the mid point of the dipole and OP is perpendicular to AB. A charge Q is placed at P where $OP = y$ and $y \gg 2a$. The charge Q experiences an electrostatic force F. If Q is now moved along the equatorial line to P' such that OP

$= \left(\frac{y}{3}\right)$, the force on Q will be close to: $\left(\frac{y}{3} \gg 2a\right)$



- (1) $3F$ (2) $\frac{F}{3}$
 (3) $9F$ (4) $27F$

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Q8

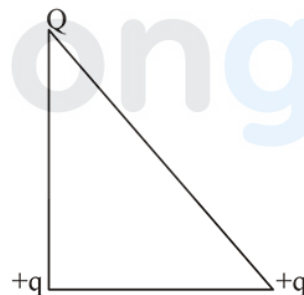
Four equal point charges Q each are placed in the xy plane at $(0, 2)$, $(4, 2)$, $(4, -2)$ and $(0, -2)$. The work required to put a fifth charge Q at the origin of the coordinate system will be:

- (1) $\frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{3}}\right)$ (2) $\frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{5}}\right)$
 (3) $\frac{Q^2}{2\sqrt{2}\pi\epsilon_0}$ (4) $\frac{Q^2}{4\pi\epsilon_0}$

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Q9

Three charges Q, $+q$ and $+q$ are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is :



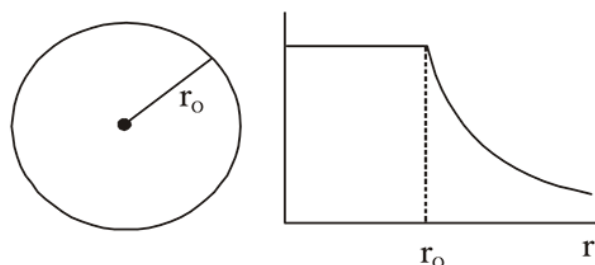
- (1) $+q$ (2) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$
 (3) $\frac{-q}{1+\sqrt{2}}$ (4) $-2q$

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Q10

Electrostatics

The given graph shows variation (with distance r from centre) of :



- (1) Electric field of a uniformly charged sphere
- (2) Potential of a uniformly charged spherical shell
- (3) Potential of a uniformly charged sphere
- (4) Electric field of a uniformly charged spherical shell

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Q11

An electric field of 1000 V/m is applied to an electric dipole at angle of 45° . The value of electric dipole moment is 10^{-29} C.m . What is the potential energy of the electric dipole?

- (1) $-20 \times 10^{-18} \text{ J}$
- (2) $-7 \times 10^{-27} \text{ J}$
- (3) $-10 \times 10^{-29} \text{ J}$
- (4) $-9 \times 10^{-20} \text{ J}$

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Q11

A particle of mass m and charge q is in an electric and magnetic field given by

$$\vec{E} = 2\hat{i} + 3\hat{j} ; \vec{B} = 4\hat{j} + 6\hat{k}.$$

The charged particle is shifted from the origin to the point $P(x = 1 ; y = 1)$ along a straight path. The magnitude of the total work done is :

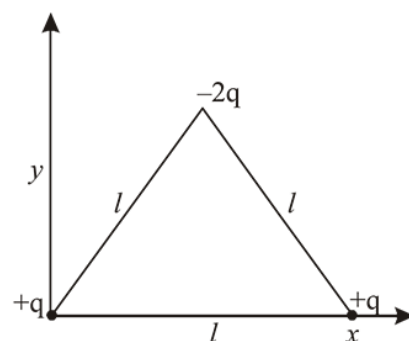
- (1) $(0.35)q$
- (2) $5q$
- (3) $(2.5)q$
- (4) $(0.15)q$

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Q12

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Determine the electric dipole moment of the system of three charges, placed on the vertices of an equilateral triangle, as shown in the figure:



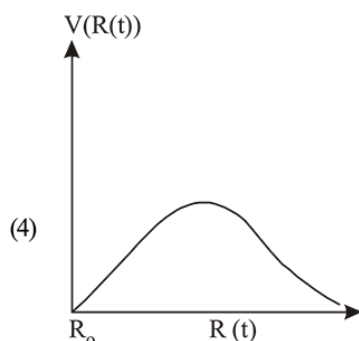
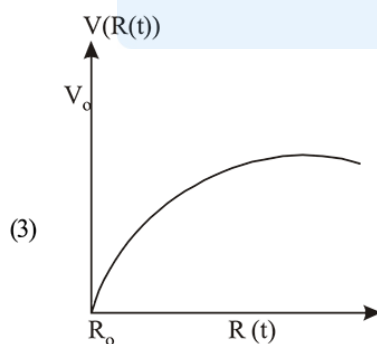
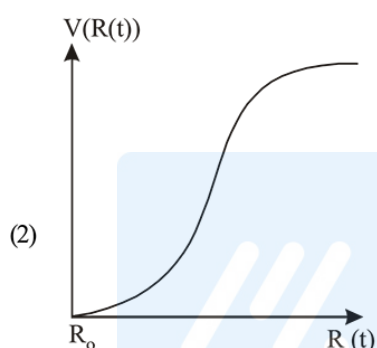
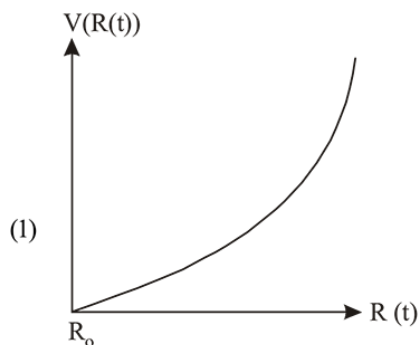
- (1) $\sqrt{3} ql \frac{\hat{j} - \hat{i}}{\sqrt{2}}$
- (2) $(ql) \frac{\hat{i} - \hat{j}}{\sqrt{2}}$
- (3) $2ql \hat{j}$
- (4) $-\sqrt{3} ql \hat{j}$

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Q13

Electrostatics

There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $V(R(t))$ of the distribution as a function of its instantaneous radius $R(t)$ is:



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Q14

The bob of a simple pendulum has mass 2 g and a charge of $5.0 \mu\text{C}$. It is at rest in a uniform horizontal electric field of intensity 2000 V/m. At equilibrium, the angle that the pendulum makes with the vertical is :

(take $g = 10 \text{ m/s}^2$)

- (1) $\tan^{-1}(2.0)$ (2) $\tan^{-1}(0.2)$
(3) $\tan^{-1}(5.0)$ (4) $\tan^{-1}(0.5)$

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Q14

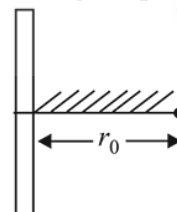
A solid conducting sphere, having a charge Q , is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V . If the shell is now given a charge of $-4Q$, the new potential difference between the same two surfaces is :

- (1) $-2V$ (2) $2V$ (3) $4V$ (4) V

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Q15

A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform density. The speed (v) of the point charge, as a function of instantaneous distance r from line charge, is proportional to :



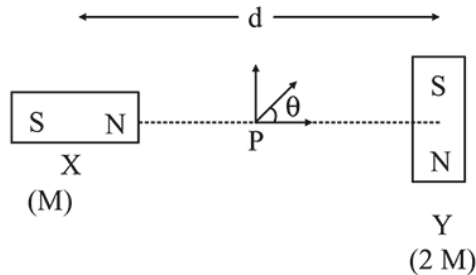
- (1) $v \propto e^{+r/r_0}$ (2) $v \propto \sqrt{\ln\left(\frac{r}{r_0}\right)}$
(3) $v \propto \ln\left(\frac{r}{r_0}\right)$ (4) $v \propto \left(\frac{r}{r_0}\right)$

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Q16

Electrostatics

Two magnetic dipoles X and Y are placed at a separation d , with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing through their midpoint P, at angle $\theta = 45^\circ$ with the horizontal line, as shown in figure. What would be the magnitude of force on the particle at that instant? (d is much larger than the dimensions of the dipole)



- (1) $\left(\frac{\mu_0}{4\pi}\right) \frac{M}{(d/2)^3} \times qv$ (2) 0
(3) $\sqrt{2} \left(\frac{\mu_0}{4\pi}\right) \frac{M}{(d/2)^3} \times qv$ (4) $\left(\frac{\mu_0}{4\pi}\right) \frac{2M}{(d/2)^3} \times qv$

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Q17

An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is :

- (1) $\sqrt{\frac{qE}{md}}$ (2) $\sqrt{\frac{2qE}{md}}$ (3) $2\sqrt{\frac{qE}{md}}$ (4) $\sqrt{\frac{qE}{2md}}$

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Q18

The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in NC^{-1} and x is in metres. The values of constants are $A = 20$ SI unit and $B = 10$ SI unit. If the potential at $x = 1$ is V_1 and that at $x = -5$ is V_2 , then $V_1 - V_2$ is :

- (1) 320 V (2) -48 V (3) 180 V (4) -520 V

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Q19

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Four point charges $-q, +q, +q$ and $-q$ are placed on y -axis at $y = -2d, y = -d, y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x -axis at $x = D$, with $D \gg d$, will behave as:

- (1) $E \propto \frac{1}{D^3}$ (2) $E \propto \frac{1}{D}$
(3) $E \propto \frac{1}{D^4}$ (4) $E \propto \frac{1}{D^2}$

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Q20

A uniformly charged ring of radius $3a$ and total charge q is placed in xy -plane centred at origin. A point charge q is moving towards the ring along the z -axis and has speed v at $z = 4a$. The minimum value of v such that it crosses the origin is :

- (1) $\sqrt{\frac{2}{m} \left(\frac{4}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$ (2) $\sqrt{\frac{2}{m} \left(\frac{1}{5} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$
(3) $\sqrt{\frac{2}{m} \left(\frac{2}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$ (4) $\sqrt{\frac{2}{m} \left(\frac{1}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

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Q21

In free space, a particle A of charge $1 \mu\text{C}$ is held fixed at a point P. Another particle B of the same charge and mass $4 \mu\text{g}$ is kept at a distance of 1 mm from P. If B is released, then its velocity at a distance of 9 mm from P is :

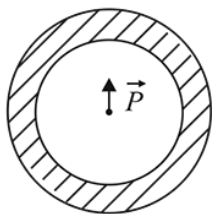
$$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2} \right]$$

- (1) 1.0 m/s (2) 3.0×10^4 m/s
(3) 2.0×10^3 m/s (4) 1.5×10^2 m/s

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Q22

Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b , and carries charge Q . At its centre is a dipole $\vec{P}$ as shown. In this case :



- (1) surface charge density on the inner surface is uniform and equal to $\frac{Q/2}{4\pi a^2}$
- (2) electric field outside the shell is the same as that of a point charge at the centre of the shell.
- (3) surface charge density on the outer surface depends on $|\vec{P}|$
- (4) surface charge density on the inner surface of the shell is zero everywhere.

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Q23

A point dipole $p = -p_o x$ kept at the origin. The potential and electric field due to this dipole on the y -axis at a distance d are, respectively : (Take $V = 0$ at infinity)

- (1) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$
- (2) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$
- (3) $0, \frac{|\vec{p}|}{4\pi\epsilon_0 d^3}$
- (4) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

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Q24

Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B , of $-Q$ each, are placed on diametrically opposite points, at equal distance, a , from the centre. If A and B do not experience any force, then :

- (1) $a = 8^{-1/4} R$
- (2) $a = \frac{3R}{2^{1/4}}$
- (3) $a = 2^{-1/4} R$
- (4) $a = R/\sqrt{3}$

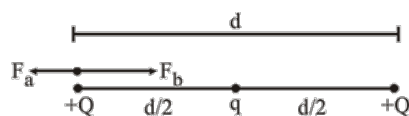
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Electrostatics - Answers

Q1

(1)



Force due to charge + Q,

$$F_a = \frac{KQQ}{d^2}$$

Force due to charge q,

$$F_b = \frac{KQq}{\left(\frac{d}{2}\right)^2}$$

For equilibrium,

$$\vec{F}_a + \vec{F}_b = 0$$

$$\Rightarrow \frac{kQQ}{d^2} + \frac{kQq}{(d/2)^2} = 0 \quad \therefore q = -\frac{Q}{4}$$

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Q3

Q2

- (2) Electric field on the axis of a ring of radius R at a distance h from the centre,

$$E = \frac{kQh}{(h^2 + R^2)^{3/2}}$$

Condition: for maximum electric field $\frac{dE}{dh} = 0$

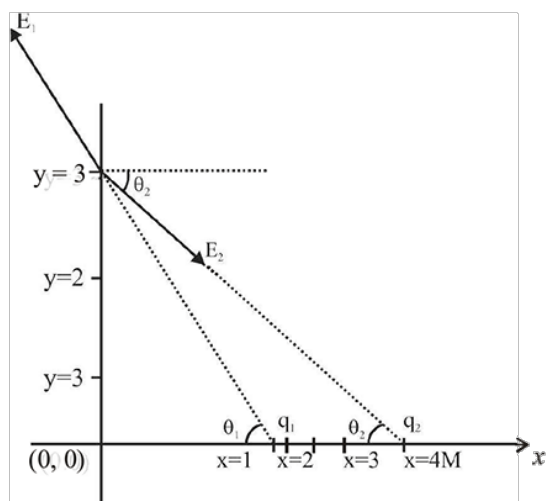
$$\Rightarrow \frac{d}{dh} \left[\frac{kQh}{(R^2 + h^2)^{3/2}} \right] = 0$$

By using the concept of maxima and minima we get,

$$h = \frac{R}{\sqrt{2}}$$

Electrostatics

(1)



Let $\vec{E}_1$ and $\vec{E}_2$ are the values of electric field due to charge, q_1 and q_2 respectively

magnitude of $E_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1^2}$

$$= \frac{1}{4\pi\epsilon_0} \frac{\sqrt{10} \times 10^{-6}}{(1^2 + 3^2)}$$

$$= (9 \times 10^9) \times \sqrt{10} \times 10^{-7}$$

$$= 9\sqrt{10} \times 10^2$$

$$\therefore \vec{E}_1 = 9\sqrt{10} \times 10^2 [\cos\theta_1(-\hat{i}) + \sin\theta_1\hat{j}]$$

$$\Rightarrow E_1 = 9 \times \sqrt{10} \times 10^2 \left[\frac{1}{\sqrt{10}}(-\hat{i}) + \frac{3}{\sqrt{10}}\hat{j} \right]$$

$$\Rightarrow E_1 = 9 \times 10^2 [-\hat{i} + 3\hat{j}] = [-9\hat{i} + 27\hat{j}] 10^2$$

Similarly, $E_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r^2}$

$$E_2 = \frac{9 \times 10^9 \times (25) \times 10^{-6}}{(4^2 + 3^2)} E_2 = 9 \times 10^3 \text{ V/m}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 (\cos\theta_2\hat{i} - \sin\theta_2\hat{j}) \therefore \tan\theta_2 = \frac{3}{4}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 \left(\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j} \right) = (72\hat{i} - 54\hat{j}) \times 10^2$$

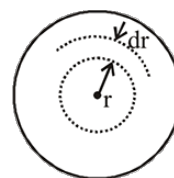
$$\therefore \vec{E} = \vec{E}_1 + \vec{E}_2 = (63\hat{i} - 27\hat{j}) \times 10^2 \text{ V/m}$$

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Q4

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(2) $Q = \int \rho dv = \int_0^R \frac{A}{r^2} e^{-2r/a} (4\pi r^2 dr)$



$$= 4\pi A \int_0^R e^{-2r/a} dr = 4\pi A \left[\frac{e^{-2r/a}}{-\frac{2}{a}} \right]_0^R$$

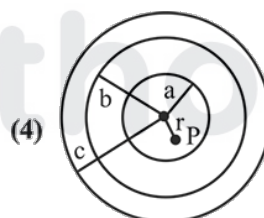
$$= 4\pi A \left(-\frac{a}{2} \right) (e^{-2R/a} - 1)$$

$$Q = 2\pi a A (1 - e^{-2R/a})$$

$$R = \frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$$

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Q5



Potential at point P, $V = \frac{kQ_a}{a} + \frac{kQ_b}{b} + \frac{kQ_c}{c}$

Since surface charge densities are equal to one another

$$\text{i.e., } \sigma_a = \sigma_b = \sigma_c$$

$$\therefore Q_a : Q_b : Q_c :: a^2 : b^2 : c^2$$

$$\therefore Q_a = \left[\frac{a^2}{a^2 + b^2 + c^2} \right] Q$$

$$Q_b = \left[\frac{b^2}{a^2 + b^2 + c^2} \right] Q, Q_c = \left[\frac{c^2}{a^2 + b^2 + c^2} \right] Q$$

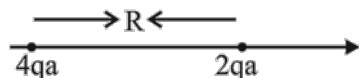
$$\therefore V = \frac{Q}{4\pi\epsilon_0} \left[\frac{(a+b+c)}{a^2 + b^2 + c^2} \right]$$

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Q6

Electrostatics

- (4) Let at a distance 'x' from point B, both the dipoles produce same potential



$$\therefore \frac{4qa}{(R+x)} = \frac{2qa}{(x^2)}$$

$$\Rightarrow \sqrt{2x} = R + x \Rightarrow x = \frac{R}{\sqrt{2}-1}$$

Therefore distance from A at which both of them produce the same potential

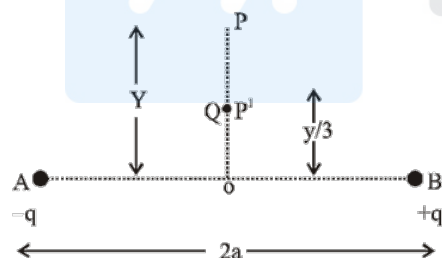
$$= \frac{R}{\sqrt{2}-1} + R = \frac{\sqrt{2}R}{\sqrt{2}-1}$$

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Q7

- (4) Electric field of equatorial plane of dipole

$$= -\frac{K\vec{P}}{r^3}$$



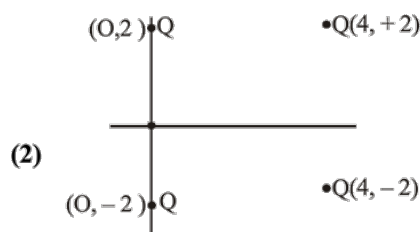
$$\therefore \text{At point P, } E = +\frac{KP}{y^3} Q$$

$$\text{At Point P}', E = +\frac{KPQ}{(y/3)^3} = 27F.$$

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Q8

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Potential at origin

$$V = \frac{KQ}{2} + \frac{KQ}{2} + \frac{KQ}{\sqrt{20}} + \frac{KQ}{\sqrt{20}}$$

$$\text{and potential at } \infty = 0 = KQ \left(1 + \frac{1}{\sqrt{5}} \right)$$

$\therefore$ Work required to put a fifth charge Q at origin $W =$

$$VQ = \frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{5}} \right)$$

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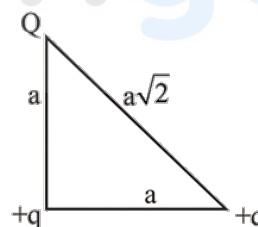
Q9

- (2) Net electrostatic energy for the system

$$U = K \left[\frac{q^2}{a} + \frac{Qq}{a} + \frac{Qq}{a\sqrt{2}} \right] = 0$$

$$\Rightarrow q = -Q \left[1 + \frac{1}{\sqrt{2}} \right]$$

$$\Rightarrow Q = \frac{-q\sqrt{2}}{\sqrt{2}+1}$$



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Q10

- (2) Electric potential is constant inside a charged spherical shell.

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Q11

Electrostatics

(2) Potential energy of a dipole is given by

$$U = -\vec{P} \cdot \vec{E}$$

$$= -PE \cos \theta$$

[Where θ = angle between dipole and perpendicular to the field]

$$= -(10^{-29})(10^3) \cos 45^\circ$$

$$= -0.707 \times 10^{-26} \text{ J}$$

$$= -7 \times 10^{-27} \text{ J}$$

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Q11

(2) Resultant force on the charged particle

$$= \vec{F}_{\text{net}} = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$= (2q\hat{i} + 3q\hat{j}) + q(\vec{v} \times \vec{B})$$

$$\text{Also } q(\vec{v} \times \vec{B}) = 0$$

So, required work done

$$W = \vec{F}_{\text{net}} \cdot \vec{S} = (2q\hat{i} + 3q\hat{j}) \cdot (\hat{i} + \hat{j})$$

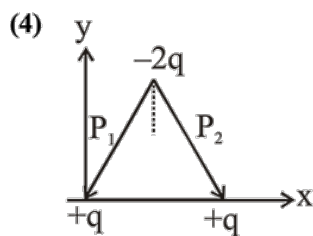
$$= 2q + 3q$$

$$= 5q$$

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Q12

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Dipole moment

= charge $\times$ distance between two charges

$$|P_1| = q(\ell)$$

$$|P_2| = q\ell$$

$$|\text{Resultant}| = 2P \cos 30^\circ$$

$$2q\ell \left(\frac{\sqrt{3}}{2} \right) = \sqrt{3} q\ell$$

$$\text{So, } \vec{P}_{\text{Resultant}} = -\sqrt{3} q\ell \hat{j}$$

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Q13

(3) Total energy of charge distribution is constant at any instant t.

$$U_f + K_f = U_i + K_i$$

$$\text{i.e., } \frac{1}{2}mV^2 + \frac{KQ^2}{2R} = 0 + \frac{KQ^2}{2R_0}$$

$$\therefore \frac{1}{2}mV^2 = \frac{KQ^2}{2R_0} - \frac{KQ^2}{2R}$$

$$\therefore V = \sqrt{\frac{2}{m} \frac{KQ^2}{2} \left(\frac{1}{R_0} - \frac{1}{R} \right)}$$

$$\therefore V = \sqrt{\frac{KQ^2}{m} \left(\frac{1}{R_0} - \frac{1}{R} \right)} = C \sqrt{\frac{1}{R_0} - \frac{1}{R}}$$

Also the slope of V – R curve will go on decreasing.

12 Jan Morning

Q14

Electrostatics

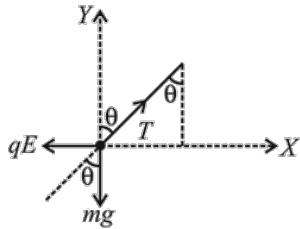
- (4) At equilibrium resultant force on bob must be zero, so

$$T \cos \theta = mg \quad \dots (i)$$

$$T \sin \theta = qE \quad \dots (ii)$$

Solving (i) and (ii) we get

$$\tan \theta = \frac{qE}{mg}$$



$$\tan \theta = \frac{5 \times 10^{-6} \times 2000}{2 \times 10^{-3} \times 10} = \frac{1}{2}$$

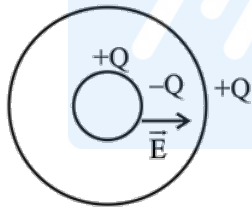
[Here, $q = 5 \times 10^{-6} \text{ C}$,
 $E = 2000 \text{ v/m}$, $m = 2 \times 10^{-3} \text{ kg}$]

$$\Rightarrow \theta = \left(\frac{1}{2} \right) = \tan^{-1}(0.5)$$

8 April Morning

Q14

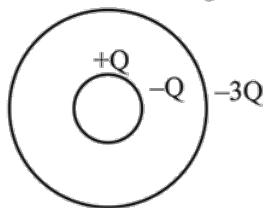
- (4) When charge Q is on inner solid conducting sphere



Electric field between spherical surface

$$\vec{E} = \frac{KQ}{r^2} \text{ So } \int \vec{E} \cdot d\vec{r} = V \text{ given}$$

Now when a charge $-4Q$ is given to hollow shell



Electric field between surface remain unchanged.

$$\vec{E} = \frac{KQ}{r^2}$$

as, field inside the hollow spherical shell = 0

$\therefore$ Potential difference between them remain unchanged

$$\text{i.e. } \int \vec{E} \cdot d\vec{r} = V$$

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Q15

- (2) Using, $[K + U]_i = [K + U]_f$

$$\text{or } 0 + Vq = \frac{1}{2}mv^2 + v'q$$

$$\text{or } \frac{1}{2}mv^2 = (V - V')q$$

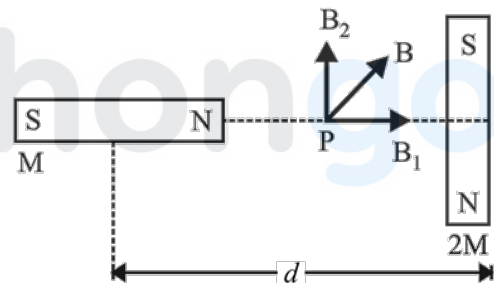
$$= -q \int_{r_0}^r E dr = q \int_{r_0}^r \frac{\lambda}{2\pi \epsilon_0 r} dr = \frac{\lambda q}{2\pi \epsilon_0} \left(\ln^3 \right)$$

$$\Rightarrow v \propto \sqrt{\ln \frac{r}{r_0}}$$

8 April Evening

Q16

$$(2) B_1 = \frac{\mu_0}{4\pi} \frac{2K}{(d/2)^3}$$



$$\text{and } B_2 = \frac{\mu_0}{4\pi} \frac{2M}{(d/2)^3}$$

$$\therefore \tan \theta = \frac{B_2}{B_1} = \frac{\frac{\mu_0}{4\pi} \frac{2M}{(d/2)^3}}{\frac{\mu_0}{4\pi} \frac{2K}{(d/2)^3}} = 1$$

$$\text{or } \theta = 45^\circ$$

The resultant field is 45° from B_1 . The angle between $\vec{B}$ and $\vec{v}$ zero, so force on the particle is zero.

8 April Evening

Q17

Electrostatics

$$(2) \quad \tau = -PE \sin \theta$$

$$\text{or } I\alpha = -PE \sin \theta$$

$$\alpha = \frac{PE}{I} (-\theta)$$

On comparing with

$$\alpha = -\omega^2 \theta$$

$$\omega = \sqrt{\frac{PE}{I}} = \sqrt{\frac{qdE}{2m\left(\frac{d}{2}\right)^2}} = \sqrt{\frac{2qE}{md}}$$

8 April Evening

Q18

$$(3) \text{ Given, } \vec{E} = (Ax + B)\hat{i}$$

$$\text{or } E = 20x + 10$$

Using $V = \int E dx$, we have

$$V_2 - V_1 = \int_{-5}^1 (20x + 10) dx = -180 \text{ V}$$

$$\text{or } V_1 - V_2 = 180 \text{ V}$$

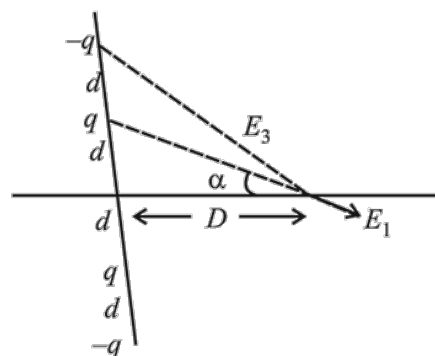
8 April Evening

Q19

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$$(4) \quad E = (E_1 + E_2) + (E_3 + E_4)$$

$$\text{or } E = 2E \cos \alpha - 2E \cos \beta$$



$$= \frac{2kq}{(D^2 + d^2)} \times \frac{D}{\sqrt{D^2 + d^2}} - \frac{2kq}{(D^2 + (2d)^2)} \times \frac{D}{\sqrt{D^2 + (2d)^2}}$$

$$= \frac{2kqD}{(D^2 + d^2)^{3/2}} - \frac{2kqD}{[D^2 + (2d)^2]^{3/2}}$$

For $d \ll D$

$$E \propto \frac{D}{D^3}$$

$$\propto \frac{1}{D^2}$$

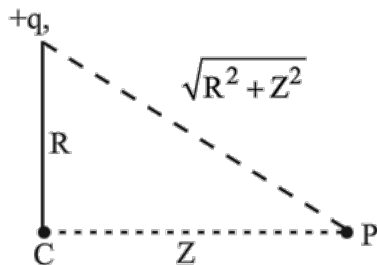
9 April Evening

Q20

Electrostatics

(3) Potential at any point of the charged ring

$$V_p = \frac{Kq}{\sqrt{R^2 + Z^2}}$$



$$R = 3a$$

$$Z = 4a$$

$$\ell = \sqrt{R^2 + Z^2} = 5a$$

The minimum velocity (v_0) should just sufficient to reach the point charge at the center, therefore

$$\frac{1}{2}mv_0^2 = q[V_C - V_P]$$

$$= q \left[\frac{Kq}{3a} - \frac{Kq}{5a} \right]$$

$$v_0^2 = \frac{4Kq^2}{15ma} = \frac{4}{15} \frac{1}{4\pi\epsilon_0} \frac{q^2}{ma}$$

$$\Rightarrow v_0 = \sqrt{\frac{2}{m} \left(\frac{2q^2}{15 \times 4\pi\epsilon_0 a} \right)^{\frac{1}{2}}}$$

10 April Morning

Q21

(3) Using conservation of energy

$$U_i = U_F + \frac{1}{2}mv^2$$

$$\frac{kq_1q_2}{r_1} = \frac{kq_1q_2}{r_2} + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}mv^2 = kq_1q_2 \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$v^2 = \frac{2kq_1q_2}{m} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$= \frac{2 \times 9 \times 10^9 \times 10^{-12}}{4 \times 10^{-6} \times 10^{-3}} \left[1 - \frac{1}{9} \right] = 4 \times 10^6$$

$$v = 2 \times 10^3 \text{ m/s}$$

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10 April Evening

Q22

(2) Surface charge density depends only due to Q . Also

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_1 \lambda}{\epsilon_0}$$

$$\text{or } E \times 4\pi r^2 = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}, r \geq R$$

12 April Morning

Q23

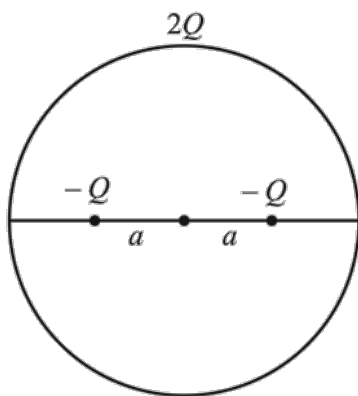
(2) The electric potential at the bisector is zero and electric field is antiparallel to the dipole moment.

$$\therefore V = 0 \text{ and } \vec{E} = \frac{1}{4\pi\epsilon_0} \left(\frac{-\vec{P}}{d^3} \right)$$

12 April Morning

Q24

$$(1) \oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$



$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int S(4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int_0^r (kr)(4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{4\pi k}{\epsilon_0} \left(\frac{r^4}{4} \right)$$

$$\therefore E = \frac{k}{4\epsilon_0} r^2 \quad \dots(i)$$

$$\text{Also } 2Q = \int_0^R (kr)(4\pi r^2) dr$$

$$= 4\pi k \left| \frac{r^4}{4} \right|_0^R$$

$$Q = \frac{\pi k R^4}{2} \quad \dots(ii)$$

From above equations,

$$E = \frac{Q r^2}{2\pi \epsilon_0 R^4} \quad \dots(iii)$$

According to given condition

$$\frac{Q^4}{4\pi \epsilon_0 (20)^2} = EQ \quad \dots(iv)$$

From equations (iii) and (iv), we have

$$a = 8^{-1/4} R.$$

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